BOUYGUES CONSTRUCTION

PPE Guide

Specifications

we life

WE LOVE LIFE.
WE PROTECT IT.

12 HEALTH & SAFETY BASICS

Our 12 Health and Safety Basics are key elements to prevent occupational accidents and protect lives. They are based on feedback and lessons learned. Encompassing the situations most frequently encountered in our construction sites or operations.
They are intented to be shared across BYCN's business and operations. They must be fully understood and obeyed by everyone.
Theses rules are intended for the people who organize work, the people who carry it out and the people who control it. They are one component of an appropriate identification and sanction policy.

Personal Protective Equipement

All personnel, including visitors, who work on or visit our construction sites shall wear as a minimum:

- Safety shoes
- A hard hat + chin strap
- High Visibility Jacket
- Safety glasses
- Hearing protection
- Safety gloves

And task-specific PPE when required as defined by task specific risk assessment
For activities in third party premises, specific PPE must be defined after task risk analysis

Preamble

Protecting the head

Protecting the hands

Protecting the feet

Protecting the eyes

Protecting the hearing

Respiratory protection

Protecting the body

Special work

Appendices

WE LOVE LIFE,
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Preamble

This Guide lists all BYCN's requirements and specifications for choosing PPE (Personal Protective Equipment).

For all special works and special cases not shown in this Guide, contact your Health, Safety and Risk-Prevention Department and DISTRIMO (for France). A risk analysis is still mandatory.

BYCN's referenced products delivered by Distrimo comply with this requirements specification.

Where you are ordering PPEs other than the referenced products, these specifications hereafter shall be the minimum requirements for the tender.

What is PPE?

It's **Personal Protective Equipment, used in addition to other measures** to eliminate or reduce risks. **Risks** means risks that are:

- *Biological (inhalation, penetration)*
- *Chemical (inhalation, penetration by dust or solvent vapour etc.)*
- *Mechanical (collisions, cuts, flying particles)*
- *Electrical (direct contact with unprotected live conductors)*
- *Thermal (working in cold or hold weather, exposure to heat, splashes of molten metal)*
- *Ionizing radiation (laser, ultraviolet, infrared)*
- *Noise*
- *Physical (vibration, etc.)*

The PPE item is chosen based on a risk assessment carried out within the business. The use of PPE is a secondary measure to protect personnel from accidental injuries or adverse health effects. **Therefore, before assigning PPE to be used, consideration shall be first given to eliminate hazards from the work or working environment by effective engineering control, or provision of safe working environment.**

The PPE item shall be:

- **Appropriate** to the risks involved
- **Suitable** for the worker
- **Compatible** with the work to be done

The wearer of the PPE must:

- **Wear the PPE items supplied**
- **Look after them and keep them in good condition**
- **Ask for replacements if they deteriorate**

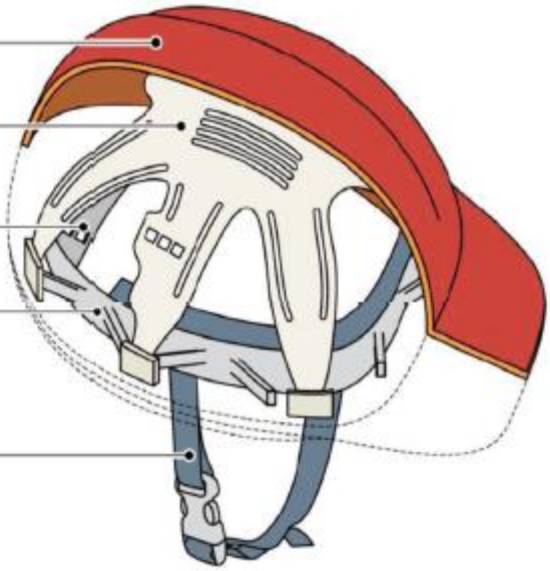
To protect myself, I wear my PPE:

At the entrance to each work site/workshop, there is a placard stating that wearing Personal Protective Equipment is compulsory: safety footwear, helmet, high-visibility clothing, hearing protection, glasses and gloves.

In addition, the risk analysis for some jobs may require specific PPE items to be worn.

Protecting the head

How does the helmet protect?



The **shell** is designed to withstand external impacts. It is made of a light and strong material such as polyethylene or polycarbonate.

The **cradle** keeps the helmet in place on the head, and acts as a shock absorber during an impact.

The **chin strap** is designed to ensure that the helmet stays on the head, particularly when working at a height.

MANDATORY REQUIREMENTS

- Shock absorbent (5 kN)
- Penetration resistant
- Flame resistant
- Anchor point for chin strap + chin strap (breaking force between 150 N and 250 N)

OTHERS REQUIREMENTS

- Strength and penetration tests at very low and very high temperatures (-20°C, -30°C, +150°C)
- Electrical insulation test
- Lateral deformation
- Splashes of molten metal

WORN AT ALL TIMES / Mandatory basic requirements:

- **Shock resistant (capable of dissipating and reducing the energy contained in an impact)**
- **Penetration resistant**
- **Flame resistant**
- **Resistant to artificial ageing**
- **Requirements relating to the constituent materials, the cradle, the chin strap and other accessories**
- **Electrically-insulated helmets (for electrical works)**
- **Chin strap**

Safety helmets are worn by everyone, including visitors, at all times on the work site or in the area of operation. Safety helmets must be protected against abrasion, scratches, cuts, punctures and paint. Any safety helmet considered to be defective, cracked, penetrated, significantly affected or exposed to extreme heat must be scrapped immediately.

Irrespective of its visual appearance, a safety helmet must be scrapped when it reaches the end of its useful life, as advised by the manufacturer or in the regulatory standards. (See the guide and/or marking under the helmet). The helmet must be cleaned regularly, and stored away from light (UV), heat and bad weather. You should avoid exposing it behind window glass, or the windscreen or rear window of a vehicle.

If the safety helmet has been subject to a violent blow, it must be replaced immediately, even if it shows no visible damage.

THE HELMET'S PERIOD OF USE AND PROPERTIES OF ITS MATERIALS

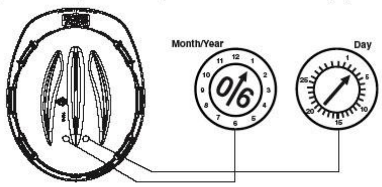
The lifetime of a protective helmet depends on its constituent materials, its use and its maintenance.

It should be inspected regularly. It must be scrapped if its condition changes (e.g. colour, damage, deformation, cracks or defibering) or after an impact. The expiry date or usage period is given in the user guide that must be supplied with each product.

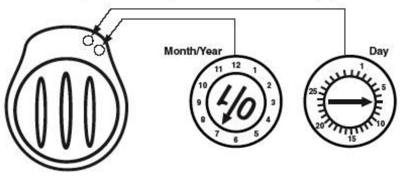
The date shown in the mandatory marking on each helmet is the date it was manufactured. Only helmets marked NF show a usage end date as shown below:

MATERIAL	POLYETHYLENE	POLYAMIDE	ABS	PHENOL TEXTILE	GLASS-FIBRE REINFORCED POLYESTER
Period of use	36 months	48 months	48 months	60 months	60 months
Resistance to ageing	Good	Good	Good	Excellent	Excellent
Resistance to UV	Acceptable	Average	Acceptable	Excellent	Excellent
Resistance to heat deformation	Up to 70°C	Up to 150°C	Up to 90°C	Up to 500°C	Up to 500°C
Melting point	150°C	220°C	180°C	Carbonization at 1000°C	Carbonization at 1000°C
Resistance to very cold conditions	Very good Limit -40°C	Average Limit -20°C	Good Limit -30°C	Excellent with no limit	Excellent with no limit
Resistance to chemicals	Good except oil and grease	Acceptable except for acids and bases	Acceptable except for acids	Good	Good

Location #1: This example shows a cap that was molded on January 3, 2006.



Location #2: This example shows a cap that was molded on July 7, 2001.



CHIN STRAP : MANDATORY

Requirement: To avoid any risk of strangulation, the chin strap or its attachment points must yield to a force of around 150-250 N. Four-point chin straps are preferred. The chin strap is not supplied with the helmet (it must be ordered in addition).

Risks: A chin strap is mandatory in all situations and in situations where the employee may lose his/her helmet without being able easily to retrieve it: for instance, during work to erect/dismantle metal frameworks and structures (scaffolding) or for work on pylons, etc.

The chin strap shall withstand a force of 50 daN for extreme work (Great heights, mountaineering).

SHOCK-PROOF HATS

PRECAUTIONS: A shock-proof hat protects against injury from a minor impact (e.g. a knock against an immobile object such as the edge of a wall). It is useful in the finishing trades, occupied sites, inside work (FM), or in cramped areas, for instance work in false ceilings. It is not designed to protect against ejected material, falling objects or suspended or moving loads. It is designed for use inside. It is made of fabric enclosed on the outside by a removable shell.

In no circumstances is the hat a substitute for a protective helmet. It may not replace the helmet for outside work. When lifting equipment is present and during work on scaffolding where there is a risk of falling objects, the helmet is necessary (the hat does not protect against such risks). The hat is designed for use in areas where there is no risk of falling items (i.e. no means of raising, no lifting equipment etc.)

MANDATORY REQUIREMENTS
- Shock absorbent
- Penetration resistant
- Anchor point for chin strap (breaking force between 150 N and 250 N)

OTHERS REQUIREMENTS
- Strength and penetration tests at very low and very high temperatures (-20°C, -30°C, +150°C)
- Electrical insulation test
- Flame resistant

The wording "Warning: this is not a protective helmet" must appear on the shell.

For work with an electrical risk, no conducting material.

VALIDATION OF ENTITY AND

RISK ANALYSIS MANDATORY

JOB/ACTIVITY	REQUIREMENTS
ALL ACTIVITIES **(structural work, sub-trades and restoration work, excluding special requirements)** **circulating within a crane's swing radius** **loading / unloading** **steering a cradle** **T&D**	Protection against knocks, falling objects, penetration (e.g. by chemicals), catching fire, artificial ageing; shock absorption Must be light (300g in the references) Must absorb perspiration (lining round the head) Easy to adjust, strong and comfortable Ergonomic (weight, comfort, bulk) Adapted to take hearing protectors such as ear muffs (Place for identification badge) Protection for the bridge of the nose Fitments: Neck protection against rain or sun: provided with a fixing system Device for fitting chin strap, and adjustable chin strap Optional integrated glasses Fitment for a head torch Drainage channels
SPECIAL CASES	
Visitors	Protection against knocks, falling objects, penetration (e.g. by chemicals), catching fire, artificial ageing; shock absorption Device for fitting chin strap + chin strap
Electrical work at voltage (HV/LV)	Electrician's helmet with integrated visor, insulating helmet
Work at a site not at voltage	General-purpose helmet + electrical insulation
Electrical work/Live working **Work related to electricity** **HV fitter**	General-purpose helmet + insulation, with (facial shield)/heat visor Chin strap with 4 attachment points, no ventilation hole
Work at a height **Assembling / Dismantling equipment (crane, formwork, huts, etc.)**	General-purpose helmet + chin strap Short visor
Workshop	General-purpose helmet or hat with shell for maintenance or handling work using the overhead travelling crane in the workshop Short visor
Jobs on occupied sites, finishing trades/after-sales service (validation of entity) **Cabling on non-live electrical equipment** **Inside work (GET/FM)**	Hat with shell protecting against knocks, falling objects and chemicals, or general-purpose helmet

Protecting the **hands**

Industrial gloves are available in a wide variety of materials, including PVC, rubber, polyethylene, leather, nylon, etc., to protect against heat, electrical shocks, grease, splinters, caustic products, acids and other harsh chemicals.

To protect against harmful chemicals, solvents, etc., gloves of special materials such as neoprene, PVC and rubber must be worn. You should select the right type of glove to protect against a harmful substance after consulting the relevant manufacturers and the Risk-Prevention Department. The gloves must be long enough to protect the wrist and forearm, where necessary, depending on the type of work to be done.

Gloves shall be discarded when they are torn, pierced or contaminated inside by hazardous materials.

Leather gloves shall be worn for electric-arc welding, gas cutting and grinding.

Insulated rubber gloves shall be used for electrical work. The gloves shall be inspected before use, as described in the manufacturer's instructions, by a competent supervisor to check for any breaks or tears in the fabric. Cotton gloves may be worn as a lining inside rubber or PVC etc. gloves, to absorb the perspiration.

The gloves shall be chosen to fit the wearer properly, so that he/she remains in full control when manipulating material and hand fatigue is reduced to a minimum.

Irrespective of the type of glove and the coverage of the anticipated risks, the levels of protection against mechanical, chemical, thermal, biological and electrical risks shall be as high as possible (involving a compromise concerning the dexterity appropriate to the job to be done). Levels shall not fall below the minima required (except in specific individual cases after approval from the competent authorities).

For work with an electrical risk, no conducting material.

The standard on **anti-cut protective gloves** has changed (ISO 13997 added):
- Better guaranteed anti-cut protection with new tests carried out during manufacture (blade pressure tested in addition to cyclical passes over the glove)
- A new marking on gloves

	Abrasion	Cutting	Tear	Puncturing	Performance letter (NEW)	Impact
E.g.	4	3	4	3	C	(P)

Cutting — *The cutting index is not necessarily indicated if the* ***performance letter*** *is given (in that case, an X would replace the number)*

Performance letter — *From A to H*. *An X indicates that the test has not been carried out (i.e. gloves that don't use the blade during the test)*

Impact — *Not indicated if not tested*

Which letter should I use?

For BYCN:
Index 5 = D or +

From 21/04/2019, performance letters must be identified on gloves. Today, some suppliers are already applying the standard (on the glove or on the label on the glove).

What should I do with old gloves?

Don't panic! Gloves that were on the market before 21 April 2019 may still be sold until 2023, and will still be valid, so you can continue to use them.

Protecting the hands

(See specifications in Appendix 1 - gloves)

MECHANICAL RISKS

Tear resistance: A (10 newtons), B (50 newtons), C (75 newtons)

Cut resistance: A (number of turns 1.2), B (number of turns 5.0), C (number of turns 20.0)

Abrasion resistance: A (100 cycles), B (2000 cycles), C (8000 cycles)

Puncture resistance: A (20 newtons), B (100 newtons), C (150 newtons)

	General						MECHANICAL					THERMAL		CHEMICAL			BIOLOGICAL	ELECTRICAL		
JOB/ACTIVITY	Tactile	Oversleeve	Dexterity	Grip	Breathability	Vibration	Abrasion	Cutting	TDM5	Tearing	Puncturing	Cold	Fire resistance.	Leak-tightness	Resistance to chemicals	Dust	Leak-tightness	Voltage resistance	Leather overglove	Insulated gloves Rubber
Concreting/major concrete constructions		X	X		X	X		B						X	X					
Preparing and using chemicals			X		X			B						X	X					
Making resin junction boxes (connecting to network)			X		X									X	X					
Painting / painting in booths														X	X					
Housework and cleaning			X		X					C				X	X					
Finishing work: delicate handling, junctions, etc. (structural work, sub-trades and restoration work)			X		X			B						X	X					
Stripping cable insulation (not live) repairing electrically-powered hand-held machines, minor stripping (Distrimo)		X	X					C												
Asbestos (and ceramic fibre) handling and lead		X	X		X			C			B			X	X					
Heavy demolition work			X		X	X	C	C		C	C			X	X					
Chopping (excluding chainsaw use), ripping, cutting (with sparking)							C	C		C	C		X			X				
Beating							C	C	X	C	C									
Pruning (chain saw)		X					C	*Class 1 (20m/s)*		C	C									
Operations on batteries at voltage					X									X				X		X

	GENERAL						MECHANICAL					THERMAL		CHEMICAL			BIOLOGICAL	ELECTRICAL		
JOB/ACTIVITY	Tactile	Oversleeve	Dexterity	Grip	Breathability	Vibration	Abrasion	Cutting	TDM5	Tearing	Puncturing	Cold	Fire resistance.	Leak-tightness	Resistance to chemicals	Dust	Leak-tightness	Voltage resistance	Leather overglove	Insulated gloves Rubber
Measuring process: work at voltage: LV and HV (+ overgloves); Connecting/disconnecting networks		X	X					C			B							X	X	X
Electro-mechanical/mechanical work not at voltage			X		X			C			B			X	X			X		
Reconditioning of electrical equipment, other non-live equipment			X		X		B	C			B			X	X			X		X
Formwork			X					C			C									
Non-live work			X					C			B							X		
Framework			X				C	C			C			X						
Delicate manipulation (electric) (not live); repairing electrical machinery			X					C			C									
Handling heavy and light items, packing, labelling, writing			X				C	C		C	C									
Work at a great height (manipulation)				X				C			B									
Assembling metal frameworks								C			C									
Manipulating portable electrical equipment						X		C	X					X		X				
Work in gloves during cold weather	X		X					C		C	C	X								
Work in polluted soil or on contaminated land			X		X			C		C	B			X	X					
Tracing (and also topography)	X		X					B												
Visitors	X							A												
Oxygen cutting		X											*100 degrees for 15 sec*							
Welding deburring			X					C			B		*100 degrees for 15 sec*							
Operations near hot spots (500 degrees for 15 sec)		X					C						*500 degrees for 15 sec*							

	GENERAL						MECHANICAL					THERMAL		CHEMICAL			BIOLOGICAL	ELECTRICAL		
JOB/ACTIVITY	Tactile	Oversleeve	Dexterity	Grip	Breathability	Vibration	Abrasion	Cutting	TDM5	Tearing	Puncturing	Cold	Fire resistance.	Leak-tightness	Resistance to chemicals	Dust	Leak-tightness	Voltage resistance	Leather overglove	Insulated gloves Rubber
Care																	X			
Work in insanitary conditions								C			C						X			
Sandblasting (put in specific)		X					C													
High-pressure washing, descaling Heavy demolition work		X							X	C	C			X						

Physical characteristics of Chemical Protection Material

Material (Designation In Matrices	Abrasion Resistance	Cut Resistance	Flexibility	Heat Resistance	Ozone Resistance	Puncture Resistance	Tear Resistance	Relative Cost
Butyl Rubber (Butyl)	F	G	G	E	E	G	G	High
Chlorinated Polythylene (CPE)	E	G	G	G	E	G	G	Low
Natural Rubber	E	E	E	F	P	E	E	Medium
Neoprene	E	E	E	G	F	E	G	Medium
Nitrile Rubber/Polyvinyl Chloride (Nitrile/PVC)	G	G	G	F	E	G	G	Medium
Polyethylene	F	F	G	F	F	P	F	Low
Polyurethane	E	G	E	G	G	G	G	High
Polyvinyl Alcohol (PVA)	F	F	P	G	E	F	G	Very High
Polyvinyl Vhloride (PVC)	G	P	F	P	E	G	G	Low
Styrene-butadiene Rubber (SBR)	E	G	G	G	F	F	F	Low
Viton	G	G	G	G	E	G	G	Very High

• Ratings are subject to variation depending on formulation, thickness, and whether the material is supported by fabric. E-excellent; G-good; F-fair; P-poor

Protecting the **feet**

Impact resistance 200J / Crushing 15 KN / Puncture resistance > 1.1 KN

Risks requiring precautions	Protective item
Mechanical risks	
Objects falling on the toes	Protective toecap
Objects falling on the metatarsals	Metatarsal protector
Objects falling on the malleoli	Malleoli protection
Crushing the end of the foot	Protective toecap
Fall with impact on the heel	Energy-absorbing heel
Fall by slipping	Non-slip sole
Walking on sharp and pointed objects	Anti-puncture insert
Walking on loose and irregular surfaces	Marked outsole lugs, high-lacing shoe
Lateral cut	Lateral cut-resistant insert
Contact with a chainsaw	Special cut-resistant upper
Electrical risks	
Electrical contact	Insulating sole
Electrostatic discharge	Dissipating sole
Electric arc	Insulating shoes
Thermal risks	
Ambient cold	Cold-resistant sole
Ambient heat	Heat-resistant sole
Heat contact	Sole resistant to contact to contact heat
Splashes of molten metal	Upper resistant to small splashes of molten metal
Firefighting	Upper and sole suitable for firefighting
Chemical risks	
Acids, bases, solvents and hydrocarbons	Resistant and water-proof upper and sole
Bad weather	
Water, snow and mud	Waterproof upper

	Safety footwear EN ISO 20345/A1	Protective footwear EN ISO 20345/A1	Occupational footwear EN ISO 20347/A1
Class I or II	SB Basic requirements	PB Basic requirements	
Class I (leather or other materials except for all-rubber and all-polymer models)	S1 Basic requirements - Closed heel - Antistatic - Energy absorption around heel S1P as for S1 + - Puncture-resistant sole	P1 Basic requirements - Closed heel - Antistatic - Energy absorption around heel	O1 Basic requirements - Closed heel - Hydrocarbon-resistant sole - Antistatic - Energy absorption around heel
	S2 as for S1 + - Impermeability to water	P2 as for P1 + - Impermeability to water	O2 as for O1 + - Impermeability to water
	S3 as for S2 + - Puncture-resistant sole - Cleated sole	P3 as for P2 + - Puncture-resistant sole - Cleated sole	O3 as for O2 + - Puncture-resistant sole - Cleated sole
Class II (all-rubber and all-polymer footwear)	S4 Basic requirements - Antistatic - Energy absorption around heel	P4 Basic requirements + - Antistatic - Energy absorption around heel	O4 Basic requirements + - Antistatic - Energy absorption around heel
	S5 as for S4 + - Puncture-resistant sole - Cleated sole	P5 as for P4 + - Puncture-resistant sole - Cleated sole	O5 as for O4 + - Puncture-resistant sole - Cleated sole

A	Antistatic
C	Conducting footwear
CI	Insulation against cold
CR	Cut-resistant upper
E	Energy-absorbing heel
HI	Insulation against heat
HRO	Sole resistant to contact heat
P	Puncture-resistant sole
FO	Hydrocarbon (fuel)-resistant sole
WRU	Upper resistant to water absorption and penetration
AN	Malleoli (ankle) protection
WR	Whole shoe waterproof
M	Metatarsal protection
I	Insulating shoe

Slip resistance requirements:

Safety footwear shall fall into one of the categories below:

SRA – *Slip resistance on ceramic tiles covered with a solution of sodium lauryl sulphate*.

SRB – *Slip resistance on a steel floor covered with glycerine.*

SRC – *Slip resistance on ceramic tiles and on a steel floor.*

JOB/ACTIVITY	REQUIREMENTS
ALL ACTIVITIES (except special cases) Also for superintending or non-operational job/supervision/ worksite management	High-top shoe Substantial (reinforced at the ends), comfortable, leakproof, easy to lace waterproof, thermally insulated, electrically insulated, breathable (+++), non-slip, flexible, puncture resistant (SR3C) including sole, good ankle support, light, sole comfortable and flexible, casing over toes (protective toecap) lateral protection against cuts strong and waterproof upper and sole high-cut upper with lacing (above the malleolus) sole insulated against cold In general, the sizes should run from 35 to 48
SPECIAL CASES	
Work in cold conditions	General-purpose shoe + padded products
Moving around an industrial store or workshop, occupied site (sales staff, supervisors and visitors etc.) outside the work site	Reinforced toe cap, anti-puncture sole (on visitors' shoes)
External visitors to the work site	PVC boots
Drivers of machinery	High-cut shoe, flexible round the ankle AND providing support when climbing down from the machinery, or high-cut shoe (if not driving permanently)
Pouring concrete, concreting work	High-top, leakproof and waterproof shoe, preventing the concrete from penetrating at all, With above-knee protection for work such as rafts or thick slabs PVC boots: chemical resistant, light, waterproof, lateral protection against cuts Protective toecap Puncture resistant Sole insulated against the cold Waterproof upper and sole
Electrical work	General-purpose shoe + no metal
All basic T&D work and travelling outside Work in the mountains, pylons	General-purpose shoe + no metal
Moving in the white room	Anti-slip overshoe
Washing (HP and no pressure)	Pressure-resistant boot for HP cleaners at least 180 bar PVC boot for non-pressurized water-jet cleaning or for cleaning with equipment under 180 bar Sole insulated against cold Waterproof upper and sole
Preparing packages, loading, unloading, reception, store logistics, after-sales service	High-cut shoe (meeting criteria for general-purpose shoe), light
Repairing portable electrical machinery as an after-sales service	High-cut shoe (meeting criteria for general-purpose shoe), light
Finishing jobs involving sustained kneeling (e.g. fixing skirting boards, laying paving, etc.)	High-cut shoe (meeting criteria for general-purpose shoe), light
Airport environments	High-cut shoe (meeting criteria for general-purpose footwear), + no metal

Everyone, including visitors, shall wear protective footwear, such as safety shoes or boots with shells protecting against impact, on the work site or in the operating plant.

The footwear shall be suitable for the work and the workplace.

- manipulating heavy objects (including manual handling)
- chemicals (storage and handling)
- extreme temperatures (heat and cold)
- slippery surfaces (steel structures, etc.)
- damp or muddy places (trenches, etc.)
- punctures and other sharp items
- electrical risks (directly and from static)

Rubber boots with or without toecaps are used to protect against splashes of corrosive chemicals.

Insulating rubber boots that are spark resistant and non conductive are used when handling explosives.

For work with an electrical risk, no conducting material.

Protecting the **eyes**

OCCUPATIONAL RISKS

The eyes shall be protected against certain occupational risks

RISKS		CONSEQUENCES FOR THE EYES
Mechanical	Impacts, dust, solid particles, grains of sand	Corneal lesions or perforations, tears in the iris, clouding of the lens
Chemical	Droplets and splashes of liquid: acids, bases, solvents and contaminating liquids	Corneal burns or disorders, acute conjunctivitis, ulcers, viral infections
Thermal	Hot liquids, molten materials, flames	Destruction of the eye
Electrical	Direct contact, electrical arc from a short-circuit	Retinal burns, corneal lesions, damage to the lens
Linked to radiation	Ultraviolet, infrared, laser, very strong light	Inflammation of the conjunctiva, cataract, retinal lesions or burns

Risks requiring precautions		Types of eye and face protectors			
		Material	Safety glasses with side shields	Goggles	Face shields
Impacts from particles ejected at high speed	*Low-energy impact (45m/s)*	Polycarbonate, Acetate	X	X	X
	Medium-energy impact (120m/s)	Polycarbonate, Acetate		X	X
	High-energy impact (190m/s)	Polycarbonate			X
	Extra tough (12m/s)	CR39, toughened glass	X	X	X
Droplets of liquid				X	
Splashes of liquid					X
Coarse dust > 5µm				X	
Gases and fine dust > 5µm				X	
Electrical arc from a short-circuit					X
Spatters of molten metal and hot solids				X	X
Gas welding			X	X	X
Arc welding					X
Ultraviolet radiation			X	X	X
Infrared radiation			X	X	X
Laser radiation			X	X	X
Solar radiation			X	X	X
Radiant heat					X

LEXICON OF MATERIALS

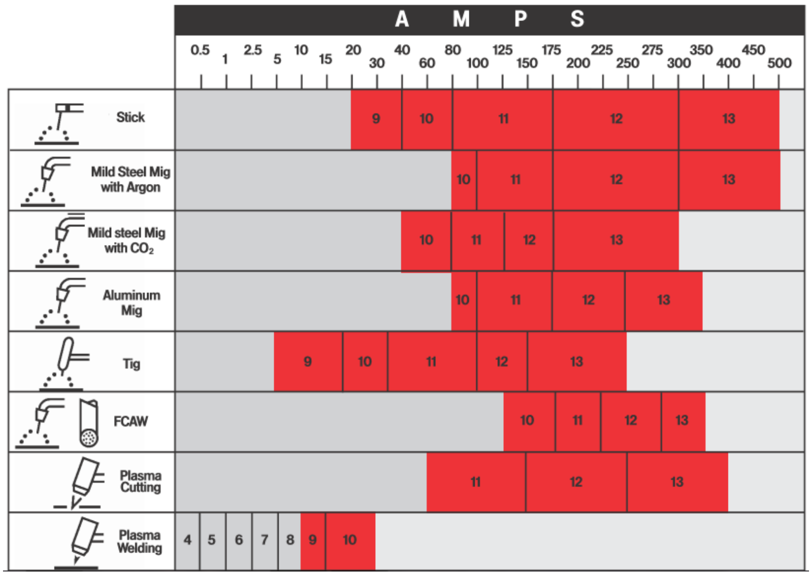
Various materials are used to make eyepieces and shields

EYEPIECES			SHIELDS	
Polycarbonate	**Acetate**	**Mineral glass**	**Polyester**	**Polyamide**
Thermoplastic and thermoformable polymer; 40 times more resistant and three times lighter than glass, provides perfect optical quality and good heat resistance (>140°C), filters UV 100%	Resin / cellulose; lighter than glass, average heat resistance (below 100°C), withstands solvents and chemicals well	Produced by fusing minerals, resists impacts, very good chemical resistance	UV filtration complete electrical insulation, withstands high temperatures well, self-extinguishing	Good thermal and electrical insulation

When cutting or grinding, at least a leakproof cover and/or a transparent face shield shall be worn with standard transparent light eye protection below.

WELDING FILTERS FOR ELECTRIC ARCS

PASSIVE WELDING FILTERS	COVER PLATES
Mineral glass for welding **Mineral glass for welding marked as heat resistant** Dim. (in mm): 105x50 – 105x30 – 110x90 Shades: 8 - 14	**Clear mineral glass** **Clear non-stick C39** Dim. (in mm): 105x50 – 105x30 – 110x90



This table indicates the recommended darkness of shade best suited to filters for different types of welding procedure. Depending on the conditions of use and the welder's preference, the figure on either side may be selected from the scale.

Welding goggles / Face shield

When arc welding or welding or cutting using oxyacetylene, welding goggles or a face shield with an optical filter lens or glass shall be used to protect against visible, infrared and ultraviolet radiation. Contact lenses may not be worn under welding goggles.

For the welding filters, the scale number refers only to the protection class. Depending on the conditions of use, you may use a filter with a number one place up or down on the scale, but it may be dangerous to use filters with too high a scale number (too dark) as this encourages the operator to approach the radiation source and breath in noxious fumes.

Scale numbers for gas welding and braze welding

	q = flow of acetylene in litres per hour			
	q<70	70<q<200	200<q<800	q>800
Welding and braze welding	4	5	6	7

Work	q = flow of acetylene in litres per hour		
	900<q<2000	2000<q<4000	4000<q<8000
Oxygen cutting	4	5	6

Chemicals

When handling hazardous materials such as corrosive or toxic chemicals, or hot liquids and fumes, chemical safety goggles shall be worn at all times to protect against injury to the eyes. They are worn in addition to any other PPE specified in the chemical’s Safety Data Sheet.

JOB/ACTIVITY	REQUIREMENTS
ALL ACTIVITIES **Entry criteria (except for special cases)**	Comfort (vision and wearability), design, side coverage, height of field of vision, adjustable to the body shape, weight, radiation (sun, etc.), water repellent, extremely scratch resistant, mist resistant, worn all the time (optical class 1 = tolerance +0.06 dioptres), impact (F, 45m/s) The levels of resistance shall be appropriate to the risks, Attention paid to the form (wide coverage) and the optical quality Eliminate glare and reflections (from water or snow) = polarized lenses The appliance has some basic resistance to everyday risks such as falling on a hard surface, ageing in light, exposure to heat or corrosion. *NB normal spectacles are not PPE and do not fulfil these requirements. Employees who wear glasses shall as a minimum be provided either with safety glasses adapted to their eyesight or overglasses. If there is a risk of flying objects, goggles shall also be worn.*
SPECIAL CASES	
Drivers of machinery	Polarized glasses, impact F
Overglasses or visitor glasses	Polycarbonate overglasses, impact F, UV2 filter
Jobs in windy places (sand)	Leakproof goggles
Jobs exposing the operator to dust	Goggles against fine dust
Grinding	Goggles & safety visor (resistant to molten metals or dust particles depending on the risk identified)
Sandblasting **Cutting** **Sanding**	Goggles, face visor or shield, highly resistant to particle impacts, comfort, leakproof
Jobs exposing the operator to splashes of hot liquids	Visor (resistant to molten metal and hot liquids)
Jobs exposing the operator to flying sparks	Goggles (worn intermittently) 120m/s (test impact)
Jobs exposing the operator to splashes of liquid, gases, fumes, aerosols and non-hazardous liquids)	Goggles and & safety visor (resistant to chemical degreasers and splashes of liquid)
Painting work **Breaking apart - flange gaskets** **Cleaning work (degreaser)**	Goggles with safety visor or visor only (resistant to chemical degreasers) (face protection)
Welding	Masque adjustable to helmet, safety glasses, high-energy impact = shields Welding goggles or a face shield with an optical filter lens or glass shall be used to protect against visible, infrared and ultraviolet radiation. The standard shade for the lens shall be provided following BS; or helmet following the risk analysis
Jobs exposing the operator to radiation and electrical arcing	Safety visor, bubble shield with protective helmet (UV2 filter), shield adaptable to the protective helmet
Live electrical connection/disconnection work	Safety visor, bubble shield with protective helmet (UV2 filter), shield adaptable to the protective helmet

Protecting the **hearing**

Hearing protectors shall be provided to each person individually: they shall not be shared between workers. The individual is responsible for looking after the hearing protector provided.

Hearing protectors shall be worn in any area where an individual has to raise his/her voice to be heard by someone standing next to him/her. For instance, near heavy equipment or noisy machinery inside workshops etc.

Note: **Choose moulded protectors where possible**. Disposable foam plugs shall be scrapped at least at the end of the day or more often if they are regularly removed and reinserted. Reusable plugs shall be washed regularly in warm soapy water. Headbands are not recommended.

The acoustic pressure shall be measured by staff who have received appropriate training in techniques for measuring sound levels using calibrated instruments. Noise surveys shall be made.

For work with an electrical risk, no conducting material.

JOB/ACTIVITY	Overall requirements
Worn all the time (following the risk assessment) **All activities**	Comfortable, suitable for wearing all the time Helmet-mounted protectors 31 db. Moulded plugs for all employees with filter corresponding the noise analysis Filter allowing through voice frequencies Resin-moulded plugs (reduces the risk of allergies)
SPECIAL CASES	
Machinery repair: operational test (acoustic panels above the workshop for staff in the vicinity)	Comfort, noise reduction (moulded plugs)
High-pressure washing at 400 bar	Noise-resistant helmet + moulded plugs or foam ear plugs
Jobs in a very noisy environment (e.g. airports and demolition sites)	Light noise-resistant helmet covering the whole ear Adaptable to the helmet Allowing moulded plugs to be worn as well or foam ear plugs No metal in helmet if necessary
Ear plugs for visitors / occasional needs	Foam ear plugs /SNR 33 dB - Reusable ear plugs - Disposable ear plugs

Respiratory protection

THE VARIOUS AIR POLLUTANTS

- **DUST AND SMOKE:** during grinding, sand blasting, sanding and crushing materials, particles of different sizes become suspended in the air.
- **MISTS:** when products are sprayed or condensed, small droplets of liquid remain suspended in the air.
- **VAPOURS:** substances in the gaseous state (liquids or solids), formed when the liquids or solids evaporate (e.g. solvents and hydrocarbons)
- **GASES:** such as air. They may be simple or complex, and spread readily at ambient temperatures. Some are very hazardous and require special protection.

TWA: TIME-WEIGHTED AVERAGE EXPOSURE

Average concentration for an 8-hour working day and a 40-hour working week. This daily exposure represents no health risk.

STEL: SHORT-TERM EXPOSURE LIMIT

Concentration below which individuals may be exposed for 15 minutes without risk to their health.

The dust mask counters dust and large particles (does not protect against gases)

- Half-face and full-face filtering masks

This type of apparatus may filter out solid aerosols, solid and liquid aerosols and gases, or gases and aerosols combined. (with cartridge appropriate to the risk, with or without dust pre-filter (e.g. painting or treatment against plant diseases). It has a facepiece covering the nose, mouth and chin, and also the eyes in the case of the full-face mask. The major part of its surface, if not the whole, is made of the filtering material. It has tabs to attach it, and in some cases one or more exhalation valves.

- Power-assisted devices

These are used in difficult working conditions (hot, long-lasting, requiring significant physical effort, etc.) The apparatus is cumbersome but very effective and comprises face protection (lining, hood, face visor, welding shield, helmet or hat) and a filter unit worn at the waist, a motor/fan and a battery. *Note*: Filtering apparatus should never be used in confined or non-ventilated spaces or in premises where the oxygen level is below a minimum of the required 17%.

Whenever workers might be exposed either to an airborne contaminant (e.g. dust, fibres, fumes, gases or other contaminants) that exceeds the occupational exposure limits defined in the legislation and/or by the company, or to an atmosphere with too much or too little oxygen, respiratory protection equipment shall be worn as specified.

The use of respiratory protection equipment shall be considered as a last resort (i.e. after all other reasonable steps have been taken, such as exhaust ventilation, using protective enclosures, working in damp conditions and using replacement materials or processes)

Breaks shall be arranged to comply with the instructions for using the mask (specific to each case, depending on the work situation and the mask used).

The use of respiratory protective equipment (RPE) should be regarded as a last resort. Only after all reasonably practicable steps, such as exhaust ventilation, protective enclosures, wet methods of work…. Wearing the mask may be recommended for some jobs in addition to suction or when working in none routine modes.

Three protection classes:

The direction from the business is to use FFP3.

The Occupational Health Department recommends the assisted ventilation mask (positive pressure) rather than non-ventilated cartridge masks (negative pressure).

	Aerosol type	Main applications	Limit
Class 1 (P1 or FFP1)	Large solid particles, not specifically toxic	Small amounts of fine dust and aqueous or oily mist, generally from sandblasting, drilling and cutting	4 x TWA
Class 2 (P2 or FFP2)	Solid and/or liquid aerosols indicated as irritants or hazards	Moderate amounts of fine dust and aqueous or oily mist, generally from plastering, cementing or sandblasting; and dust from soft wood	10 x TWA
Class 3 (P3 or FFP3)	Toxic solid and/or liquid aerosols	Large amounts of fine dust and aqueous or oily mist, generally from handling toxic powders present in the pharmaceutical industry or from jobs involving biological agents and fibres	With half-face mask: 50 x TWA With full-face mask: 200 x TWA

To select the right protection, you must take into account of:
- **the precise nature of the pollutant** (dust, smoke, bacteria, gas or fumes) **and its concentration,**
- **the planned time to be spent** in the polluted atmosphere,
- **the difficulty of the work** to be done,
- **the configuration of the premises** where the work will be done (confined or ventilated).

The apparatus must be cleaned and disinfected after each use (unless it is part of personal equipment) and prepared for subsequent reuse or storage.

The devices will not be cleaned and maintained except using the products indicated by the manufacturer, otherwise their operation may be impaired or they may be damaged.

JOB/ACTIVITY	REQUIREMENTS
GENERAL	Dust mask FFP3 (recommendation use max 15 min) Dust large-particle or mask: filtering half-or full-face filtering mask (>15min) Assisted-ventilation mask (>1h) Comfort: light, easy to adjust to the body shape, provides good visibility, enables the helmet or hat to be worn, ventilation eliminating misting if visor used or safety glasses
SPECIAL CASES	
Tapping, sanding, sweeping, cutting wood, grinding Work in an environment with cement or silica dust	Ideally the FFP3 half-face mask with cartridge and the assisted-ventilation mask. Mask to be chosen based on the risk analysis. Does not mist up safety goggles Easy to clean Comfortable, easy to adjust Good performance during the period of use + easy to breathe through the filters General Local exhaust ventilation whenever possible.
Electrical work on air coolers	Half-face mask or all-vision face piece with A2P3 filter cartridge
Work on electrical equipment with ceramic fibre present	Half-face mask or all-vision face piece with A2P3 filter cartridge
Hydrocarbon vapours	Half-face mask or all-vision face piece with A2P3 filter cartridge
Painting work outside booths	Mask depending on the chemical risk analysis carried out for each structure.
Use of chemicals	Mask depending on the chemical risk analysis carried out for each structure.
Painting work in booths	Comfort: light, easy to adjust to body shape, provides good visibility, enables helmet or hat to be worn, ventilation eliminating misting if visor used.
Preparation before painting	Comfort: light, easy to adjust to body shape, provides good visibility, enables helmet or hat to be worn, ventilation eliminating misting if visor used.
Painting	Cartridge mask, half- or full-face
Work in confined areas	Cartridge mask, half- or full-face
Grinding in s	Assisted ventilation with filtering
Cleaning machinery (after-sales service)	P3 filtration
De-scaling work, lead-based painting	Cartridge mask, half- or full-face
Welding, laboratory evacuation	Mask depending on the chemical risk analysis carried out for each structure. small FFP3 Do not impede breathing or demand extra effort to breathe in or out No misting
Torch welding/welding	Electronic soldering mask and assisted-ventilation mask for enclosed areas

Main criteria to take into account when choosing suitable breathing apparatus

PARAMETER	EFFECT	RECOMMENDATION
Physical characteristics of the wearer	Facial features such as scars, unshaven hair (last shave at least eight hours ago), sideburns or skin rashes that could compromise the protection from a half- or full-face mask, as the irregular surface would prevent the face piece remaining leak tight. A single face piece may not suit all face shapes. Some types of equipment may be very heavy, preventing some people from wearing it. There may be allergies to some types of materials directly in contact with the skin.	If so, helmets or hoods should be selected in preference, if they provide the minimum required level of protection. When choosing the mask, it is important to consult the supplier about the different sizes available. Users shall be offered a selection so that each person can select the model that is best suited to him/her. Select another type of device offering the same level of protection. Select another type of device offering the same level of protection.
Glasses	Glasses are not compatible with a full-face mask, or even with some helmets or hoods, because the side-pieces prevent the face piece from being leakproof.	There are special devices that attach corrective lenses directly to the inside of a full-face mask. Otherwise, respiratory protection equipment with a hood compatible with wearing glasses may be used, if it has the necessary degree of protection.
Contact lenses	When the user wears respiratory protection equipment, the lenses may move around the eye, or the air flow may cause the eyes to become excessively dry. In either case, the wearer may be tempted to remove his/her mask to remedy the problem.	You should therefore assess how easy it is for the wearer to reach a non-polluted area to remove the equipment. Otherwise, if the mask cannot be removed quickly without spreading pollution, it may be necessary to advise that contact lenses are nor worn.
Accessories	Accessories (necklaces, earrings, scarves, turbans, headgear or body piercings etc.) shall not compromise the leakproofness of the face piece, nor reduce the normal air flow during use.	Where such items cannot be removed, respiratory protection equipment with an adapted hood model may be used, if it has the right level of protection.
Tools used	The tools used at the workstation shall not interfere with respiratory protection. For instance, if: - The tools or processes produce electrical or magnetic fields - There is a risk of flying molten particles (from welding, cutting etc.) - Some tools may transmit vibrations or issue jets of air or particles that could affect the mask's leakproofness.	- Respiratory protection equipment that are not sensitive to these fields shall be selected - Equipment with the highest heat and flame resistance shall be selected - The operating procedures shall be modified to avoid any harmful interference with the mask
Interaction with other personal protective equipment	In situations with multiple risks, the different items of personal protective equipment shall be compatible with each other so the protection is not compromised.	For some applications such as welding or shot-blasting, there is integrated equipment including protection for the head, eyes, ears and respiratory tract.

Selecting breathing apparatus

It is essential that a designated competent person (risk prevention in consultation with occupational health) selects the most appropriate type of breathing apparatus. The choice depends on a variety of factors.

Among the many to consider when choosing appropriate respiratory protection equipment for a given situation involving polluted air, the following are worth quoting:

- characteristics of the operation or process
- nature of hazard
- oxygen concentration at the location
- period of protection
- field of vision
- communication
- restriction of workplaces
- characteristics and limitations of the respirators
- user's medical condition
- adjustability (comfort, lightweight, etc.) and face seal
- availability
- decontaminating and storing the equipment

Selecting body protection

Take into account:
- *Type and provenance of the hazards* (chemicals, extreme temperatures, electrical risks, radiation risks, etc.)

- *Probability of dermal exposure* (probability, period of exposure, etc.)

- *Consequence of direct contact with the skin* (type of lesion, severity, etc.)

- *Protective measures* (estimate of the level of protection, whole body, parts of the body, etc.)

Working groups are currently studying this subject: you will find below recommendations until the final conclusions are available. Special cases shall be taken into account.

JOB/ACTIVITY	Overall requirements
All work	Fluorescent material that contrasts well with the environment (yellow, orange, red) Retro-reflective material (reflects the light from lamps and projectors) High visibility (orange and yellow colours) class 2 Work trousers & jacket / fluorescent waistcoat / parkas => Resistant to static, acids, fire, mechanical stresses (tearing), bad weather & cold HV waistcoats - Class 2 multi-pockets - half-mesh, no pockets Long-sleeved tee-shirt (short-sleeved could be banned following the risk analysis) Class 2 and class 3 if working near traffic areas visibility Retro-reflective but sill comfortable (+++), breathing (+++), several sizes (+++), not inflammable , UV protection, water repellent, light Elasticated adjustment at both waist and ankles Integrated knee protection, adjustable to the correct position to avoid the knee pads slipping Tear resistant Different sizes of pockets in the jacket Jacket with adjustable cuff Fleece for winter, removable for summer Cold-resistant parka 4 in 1 parka
SPECIAL CASES	
Welding/torch welding work	Fire-retardant clothing Welder's apron (resistant to heat + flying sparks) Does not propagate flame Oversleeves, gaiters Like ®Proban clothing for boilermakers and more generally for employees whose work consists of welding and grinding. + Leather protection
Electrical cutting /live working	Does not propagate flame
Electrician, HV workstation	Trousers, jacket, long-sleeved tee shirt: Nomex fabric
Chopping wood	Special fabric Over-trousers

Cutting metal	Over-trousers (no apron) + jacket
Electrical work	No metal buttons or fastenings
Painting in booths	Disposable coveralls worn over working clothes.
Sandblasting in booths	One-piece leather coveralls.
Laboratory, insanitary conditions, etc.	Leakproof coveralls
Nuclear environment	Protection against contaminated dust
Joinery	Velcro-type band on the thigh to hold a cloth/tissue for cleaning hand tools that is easy to change (silicone)

This paragraph deals only with clothing protecting against chemicals:

The kit may include gloves, dungarees, safety glasses, aprons and boots, depending on the work to be done.

The purpose of the clothing is to prevent chemicals coming into contact with the skin when handling hazardous chemicals, and especially when distributing chemicals or solvents. When selecting body protection that satisfies the requirements, the person responsible shall consult the appropriate manufacturers or receive information about their products. The proposed selection shall be submitted to the Risk-Prevention Department and to Distrimo for approval before use.

The clothing shall be inspected visually before use, since the hazardous chemicals that encroach via the surface of the protective clothing may cause serious injuries or dermatitis. These inspections shall include a close visual inspection to identify signs of deterioration in the materials, colour changes, cracking, brittleness, etc. Special attention shall be paid to the seams.

Electrical risk : There si a standard for Electrical insulating protective clothing for low-voltage installations NF EN 50286

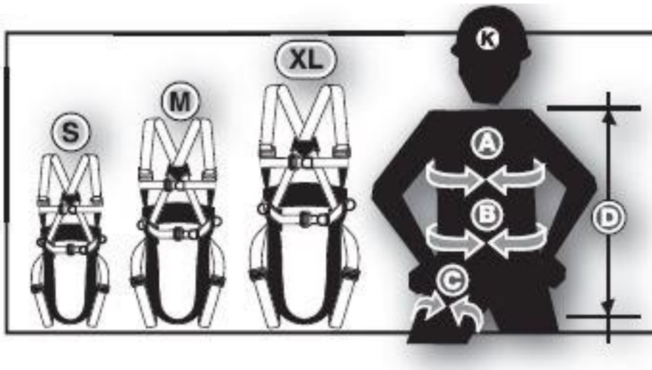
FOR PROTECTION AGAINST FALLS

Equipment and accessories to be defined depending on the work situation (cradle, unloading bay, flat roof, etc.). Contact your Risk-Prevention Department. To use a harness, I must be trained and authorized. The appropriate equipment shall be identified based on the risk analysis for the work situation.

For accessories related to working at a height, you should consult the Risk-Prevention Department (karabiner, runners, energy absorber, descending rings, rigger’s bag, emergency evacuation kit)

For work with an electrical risk, no conducting material.

Guide to sizes



	S	M	XL
A	60 - 100	80 - 120	100 - 160
B	75 - 100	90 - 110	100 - 125
C	45 - 80	45 - 100	45 - 120
D	55 - 65	65 - 80	80 - 90

JOB/ACTIVITY	REQUIREMENTS
Work involving manoeuvring electrical equipment or making connections to a distribution frame at voltage or installing collective insulation protection	Arc flash coveralls (resistance according to the energy levels in customers' electrical switchboards)
Electrical work	Non-conducting material
Maritime work	Life vest Characteristics to use to determine the type of vest: Buoyancy index of at least 150 newtons to ensure that an unconscious person turns over and floats (in extreme conditions, 275N), Self-inflating
Proximity to water	Life vest (manual or automatic) with buoyancy of at least 150N resistant to flying incandescent particles buoy + lifeline Thigh boots
Confined spaces	Mask with 0_2 supply Gas detector
ATEX	Anti-static equipment
Very bright areas	Protect eyes against glare
Welding at height using a harness	Harness + lanyard with fire protection
Workshop, mechanical	Fire-resistant clothing
Chain sawing	Gaiters
Welder boilermaker	General-purpose shoes + leather overshoe or lace protectors or leather boots All-leather gaiters
Welding work	Leather hood for full-time welders or general-purpose helmet + special welding mask
Welding/torch welding work	Fire-retardant clothing Welder's apron (resistant to heat + flying sparks) Does not propagate flame Oversleeves, gaiters Proban clothing for boilermakers and more generally for employees whose work consists of welding and grinding. + Leather protection
Welding	Masque adjustable to helmet, safety glasses, high-energy impact = shields Welding goggles or a face shield with an optical filter lens or glass shall be used to protect against visible, infrared and ultraviolet radiation. The standard shade for the lens shall be provided following BS, or helmet following the risk analysis
Torch welding	Welding mask with safety helmet

JOB/ACTIVITY	REQUIREMENTS
Asbestos and lead **Work in an area where hazardous products are present** **Breaking apart gaskets - equipment on flange**	Follow the regulatory framework plus comfort, able to be adjusted and to breathe but proof against asbestos fibres Kit for working with asbestos and lead to be made up: - Disposable single-use coveralls: comfortable, the right size for the operator, proof against fibres, perspiration resistant o Asbestos: category 3 type 5/6 with hood = proof against particles o Lead: category 3 type 4 - 6 with hood in the case of fogging o Ceramic fibre: category 3 type 5/6 with hood - American Army adhesive fabric tape: o Make leakproof at the wrists and seams coveralls/shoes and hood/ respiratory protection equipment - Overboots: non-tear, if possible with a non-slip sole o To be preferred, but be aware of the risk of slipping on wet floors. If overboots are not worn, the only reason shall be that safety shoes designed exclusively for work in contaminated areas are worn. - Respiratory protection equipment with unforced ventilation: Comfortable, easy to adjust, leakproof as far as possible based on the operator's body shape, small and light filters (cartridges) o Lead: § Half-face cartridge mask, type P3. This PPE item requires wearing goggles (see eye protection) § Full-face cartridge mask, type P3 § Mask with filters suitable for A2P3 for torch welding leaded iron o Asbestos/ceramic fibre: § Full-face cartridge mask, type P3 - Respiratory protection equipment with assisted ventilation: Comfortable, easy to adjust, leakproof as far as possible based on the operator's body shape, small and light filters (cartridges) either under placed the mask with the batteries at the waist or with both batteries + filters at the waist. Maximum weight of the assembly o Lead: § Half-face mask with assisted ventilation and P3 filters. This PPE item requires wearing goggles (see eye protection) § Full-face cartridge mask with assisted ventilation, P3 filters § Assisted ventilation mask with A2P3 filters for torch welding leaded iron. o Asbestos/ceramic fibres: § Full-face cartridge mask with assisted ventilation, P3 filters
Harness kit	Structural work, cradles and after-sales service: Tractel-type harness Short lanyards for working on a cradle (lanyard < 2m) 2m lanyards (after-sales service, structural work) Stops fall in 1m (erecting scaffolding and cradles, etc.) Quick Lock attachment system
Work at a height: Intermittent work with harness (after-sales service, making safe)	Adjustable straps (at least shoulder, leg and chest straps) across the whole body, and at least one dorsal anchor point Fall safety set, energy absorber? If the anchor point is located above the attachment point on the harness or at the same level: fall safety set; if the anchor point is below the attachment point: energy absorber integrated into the lanyard

Work at a height, **Work in a cradle (installing poles, keying beams, etc.)** **Assembling equipment work at a height** **Erecting huts**	Light, robust, practical Harness with at least 2 attachment points Lanyard, helmet and chin strap Harness adjustable to helmet worn with chin strap Easy to adjust/adjustable (+++) Comfortable leg straps Not damaged if exposed to solvents or chemicals Inflammable Attachment: 1 attachment point since basket in the cradle
Use of extension arms for loading and unloading.	Waistcoat-type harness, easy to put on and adjust
Suspended platforms **Situation: platform fixed at the top of the building**	Attachment: to anchor point (or temporary life line)
Exceptional situations (precast slabs) **Situation: laying precast slabs**	Attachment: to anchor point (Quicklock or runner)
Flat roof **Situation: flat room with no collective civil engineering**	Attachment: to life line
EHV pylon erectors	Harness with 4 points (5th point removed), comfort, positions worker at workstation Working while suspended is not permitted Lanyard to position at workstation With triple-action connectors (karabiner): no Manucroche Y lanyard with energy absorber With triple-action connectors (karabiner): no Manucroche Lanyard to position at workstation with steel core With triple-action connectors (karabiner): no Manucroche 9m anchor lines (diam 11mm) With triple-action connectors (karabiner): no Manucroche Stick-run Life line (diam 11) + runner Automatic winder with short (2m) cable that works in all directions
Crane erector **Huts (with anchoring on hut)**	HARNESS EN 361/358 WITH DORSAL ANCHOR, STERNAL ANCHOR, ANCHOR TO MAINTAIN AT WORK AND GEAR CARRIER LANYARD WITH EN 355 SHOCK ABSORBER LENGTH 1.5 m; WITH CONNECTOR 1 WIDE OPENING (AUTOMATIC) AND CONNECTOR 2 MULTIDIRECTIONAL PEAR-SHAPED SCREW-IN OR AUTOMATIC (except for hut, if attachment possible elsewhere) LANYARD TO POSITION AT WORK Length 2 m - EN 358 IN ADDITION TO THE FALL ARREST SYSTEM RUNNER EN 795 Class B Length 1,5 m IN ADDITION TO THE EXISTING ANCHORS (APPLIES IN ANY SITUATION) STEEL CONNECTORS EN 362 CONVENTIONAL SHAPE WITH 1/4-TURN AUTOMATIC LOCKING AUTOMATICALLY-TRIGGERED CABLE FALL ARRESTOR EN 360 A CABLE Length 10 m if attachment not on hut

Erecting metal frameworks	HARNESS EN 361/358 WITH DORSAL ANCHOR, STERNAL ANCHOR, ANCHOR TO MAINTAIN AT WORK AND GEAR CARRIER LANYARD WITH EN 355 SHOCK ABSORBER LENGTH 1.5 m; WITH CONNECTOR 1 WIDE OPENING (AUTOMATIC) AND CONNECTOR 2 MULTIDIRECTIONAL PEAR-SHAPED SCREW-IN OR AUTOMATIC AUTOMATICALLY-TRIGGERED FALL ARRESTOR EN 360 Length 1.2 – 1.5 m CONNECTOR 1 WIDE OPENING (AUTOMATIC) AND CONNECTOR 2 MULTIDIRECTIONAL PEAR-SHAPED SCREW-IN OR AUTOMATIC AUTOMATICALLY-TRIGGERED CABLE FALL ARRESTOR EN 360 A CABLE Length 10m LANYARD TO POSITION AT WORK Length 2 m - EN 358 IN ADDITION TO THE FALL ARREST SYSTEM RUNNER EN 795 Class B Length 1.5 m IN ADDITION TO THE EXISTING ANCHORS (APPLIES IN ANY SITUATION) STEEL CONNECTORS EN 362 CONVENTIONAL SHAPE WITH 1/4-TURN AUTOMATIC LOCKING
Electrical Installations	HARNESS EN 361/358 WITH DORSAL ANCHOR, STERNAL ANCHOR, ANCHOR TO MAINTAIN AT WORK AND GEAR CARRIER LANYARD WITH EN 355 SHOCK ABSORBER LENGTH 1.5 m; WITH CONNECTOR 1 WIDE OPENING (AUTOMATIC) AND CONNECTOR 2 MULTIDIRECTIONAL PEAR-SHAPED SCREW-IN OR AUTOMATIC AUTOMATICALLY-TRIGGERED FALL ARRESTOR EN 360 Length 1.2 – 1.5 m CONNECTOR 1 WIDE OPENING (AUTOMATIC) AND CONNECTOR 2 MULTIDIRECTIONAL PEAR-SHAPED SCREW-IN OR AUTOMATIC LANYARD TO POSITION AT WORK Length 2 m - EN 358 IN ADDITION TO THE FALL ARREST SYSTEM RUNNER EN 795 Class B Length 1.5 m IN ADDITION TO THE EXISTING ANCHORS (APPLIES IN ANY SITUATION) STEEL CONNECTORS EN 362 CONVENTIONAL SHAPE WITH 1/4-TURN AUTOMATIC LOCKING

Heavy-goods driver	HARNESS EN 361/358 WITH DORSAL ANCHOR, STERNAL ANCHOR, ANCHOR TO MAINTAIN AT WORK AND GEAR CARRIER LANYARD WITH EN 355 SHOCK ABSORBER LENGTH 1.5 m; WITH CONNECTOR 1 WIDE OPENING (AUTOMATIC) AND CONNECTOR 2 MULTIDIRECTIONAL PEAR-SHAPED SCREW-IN OR AUTOMATIC AUTOMATICALLY-TRIGGERED FALL ARRESTOR EN 360 Length 1.2 – 1.5 m CONNECTOR 1 WIDE OPENING (AUTOMATIC) AND CONNECTOR 2 MULTIDIRECTIONAL PEAR-SHAPED SCREW-IN OR AUTOMATIC AUTOMATICALLY-TRIGGERED CABLE FALL ARRESTOR EN 360 A CABLE Length 10m LANYARD TO POSITION AT WORK Length 2 m - EN 358 IN ADDITION TO THE FALL ARREST SYSTEM RUNNER EN 795 Class B Length 1.5 m IN ADDITION TO THE EXISTING ANCHORS (APPLIES IN ANY SITUATION) STEEL CONNECTORS EN 362 CONVENTIONAL SHAPE WITH 1/4-TURN AUTOMATIC LOCKING
Formwork erector	HARNESS EN 361/358 WITH DORSAL ANCHOR, STERNAL ANCHOR, ANCHOR TO MAINTAIN AT WORK AND GEAR CARRIER LANYARD WITH EN 355 SHOCK ABSORBER LENGTH 1.5 m; WITH CONNECTOR 1 WIDE OPENING (AUTOMATIC) AND CONNECTOR 2 MULTIDIRECTIONAL PEAR-SHAPED SCREW-IN OR AUTOMATIC LANYARD TO POSITION AT WORK Length 2 m - EN 358 IN ADDITION TO THE FALL ARREST SYSTEM RUNNER EN 795 Class B Length 1.5m IN ADDITION TO THE EXISTING ANCHORS (APPLIES IN ANY SITUATION) STEEL CONNECTORS EN 362 CONVENTIONAL SHAPE WITH 1/4-TURN AUTOMATIC LOCKING

APPENDIX 1 – Glove specifications

Properties of the materials used

Latex (NR): supple, resists chemicals
Nitrile (NBR) : resists perforation, oils and solvents
Neoprene: resists heat, abrasion and chemicals
Butyl: Resists chemicals
PVC (Polyvinyl chloride): resists abrasion, waterproofs
PVA: resists chemicals
Polyurethane: resists abrasion and tearing; good dexterity and tactile sensitivity

Mechanical risks	Tear resistance: A (10 newtons), B (50 newtons), C (75 newtons) Cut resistance: A (1.2 no. turns), B (5.0 no. turns), C (20.0 no. turns) Abrasion resistance: A (100 cycles), B (2000 cycles), C (8000 cycles) Puncture resistance: A (20 newtons), B (100 newtons), C (150 newtons)
Mechanical risks	Cut resistance (DTM5)
Chemical and microbiological risk	**Enhanced chemical protection: determination of permeability resistance** If a glove does not leak and reached a performance level of at least 2 (>30 minutes) in the permeation test for at least three of the listed products, it shall bear the pictogram “Specific chemical hazards” with the letters corresponding to the chemicals for which at least level 2 was reached. There may be up to 12 letters.
Chain sawing	Class 1 - the protective surface must be at least 110 mm wide and 120 mm high at the highest point (without the fingers). To protect the back of the hand and the four fingers, the protective surface must be at least 110 mm wide and 190 mm high from the base to the highest point

Code letter	Chemical
A	Methanol
B	Acetone
C	Acetonitrile
D	Dichloromethane
E	Carbon disulphide
F	Toluene
G	Diethylamine
H	Tetrahydrofuran
I	Ethyl acetate
J	n-Heptane
K	Caustic soda 40%
L	Sulphuric acid 96%

Heat and fire risk

Performance level	1	2	3	4
A Resistance to flammability (time in seconds after flame and after glow time)	<20s no requirement	<10s <120s	<3s <25s	<2s <5s
B Contact heat (contact temperature for a minimum period of 15s)	100°C	250°C	350°C	500°C
C Convective heat (heat transfer delay)	>4s	>7s	>10s	>18s
D Radiant heat (heat transfer delay)	>7s	>20s	>50s	>95s
E Small drops of molten metal (number of drops)	>10	>15	>25	>35
F Large quantities of molten metal (mass)	30g	60g	120g	200g

Electrical risks

Class	Maximum thickness (mm)	Test voltage (volts)	Usage voltage (volts)
00	0.5	2500	500
0	1	5000	1000
1	1.5	10000	7500
2	2.3	20000	17000
3	2.9	30000	26500

Chemical resistance

The chemical resistance of protective gloves is defined based on the following characteristics (the performance levels are classed in increasing order):

◦ Resistance to leaks (3 levels)

◦ Permeation resistance (6 levels)

Protection against ionizing radiation and radioactive contamination

The leakproof gloves (pictogram) used within contained areas against radioactive contamination, are highly resistant to permeation and water vapour and to ionizing radiation. They shall contain a defined quantity of lead, shown on the gloves.

Cold

Performance level	1	2	3	4
A Convective cold, thermal insulation: I_{TR} in m^2K/W	$0.10<I_{TR}<0.15$	$0.15<I_{TR}<0.22$	$0.22<I_{TR}<0.30$	
B Contact cold, thermal resistance R in m^2K/W	$0.025\leq R<0.050$	$0.050\leq R<0.100$	$0.100\leq R<0.150$	$0.150\leq R$
C Permeability to water (for at least 30 min)	Successful			

The standard on **anti-cut protective gloves** has changed (ISO 13997 added):

- Better guaranteed anti-cut protection with new tests carried out during manufacture (blade pressure tested in addition to cyclical passes over the glove)
- A new marking on gloves

	Abrasion		Tear		Performance letter NEW	
E.g.	4	3	4	3	C	(P)
		Cutting *The cutting index is not necessarily indicated if the performance letter is given (in that case, an X would replace the number)*		Puncturing	*From A to H* *An X indicates that the test has not been carried out (i.e. gloves that don't use the blade during the test)*	Impact *Not indicated if not tested*

Which letter should I use?

For BYCN:
Index 5 = D or +

From 21/04/2019, performance letters must be identified on gloves. Today, some suppliers are already applying the standard (on the glove or on the label on the glove).

What should I do with old gloves?

Don't panic! Gloves that were on the market before 21 April 2019 may still be sold until 2023, and will still be valid, so you can continue to use them.

GUIDE TO CHEMICAL RESISTANCE

This table provides only general information about the materials. You should bear in mind that a glove's resistance is influenced by factors such as the exact nature of the chemical, its temperature, its concentration, the thickness of the glove and the immersion time, etc.
We advise you to refer to the information on the chemical resistance of each glove and to carry out a preliminary test to determine if the glove is suitable for use in the working conditions.

Key: P=Poor, F=Fair, G=Good, E=Excellent, NR=Not Recommended

Chemical	Neoprene	Nitrile	Latex	PVC
Acetaldehyde	E	P	F	NR
Acetic Acid	E	G	G	F
Acetone	G	NR	G	NR
Acetonitrile	F	NR	F	NR
Ammonium Hydroxide<30%	E	E	G	E
Amyle Acetate	NR	E	F	P
Amyl Alcohol	P	G	G	NR
Aniline	G	NR	P	F
Animal Fats	E	E	P	G
Battery Acids	E	E	G	E
Benzaldehyde	NR	NR	F	NR
Benzene	NR	P	NR	NR
Benzoly Chloride	NR	NR	P	NR
Butane	F	E	P	P
Butyl Acetate	NR	F	P	NR
Butyl Alcohol	E	P	E	G
Butyl Cellusolve*	E	E	E	NR
Carbon Acid	E	P	P	G
Carbon Disulfide	NR	NR	NR	NR
Carbon Tetrachloride	P	G	NR	NR
Castor Oil	E	E	E	E
Cellosole Acetate	E	G	G	NR
Cellosole Solvent	E	G	E	NR
Chlorobenzene	NR	NR	NR	NR
Chloroform	F	F	NR	NR
Chloronaphalens	NR	F	NR	NR
Chloroethene VG	NR	F	NR	P
Chromic Acid	F	F	NR	G
Citric Acid	E	E	E	E
Cottonseed Oil	E	E	P	G
Cresols	G	G	P	F
Cutting Oil	E	E	F	P
Cyclohexane	F	E	P	P
Cyclohexanol	E	E	P	G
Dibutyl Phthalate	F	G	P	G
Diethylamie	P	F	NR	NR
Di-Isobutyl Ketone	P	E	P	P
Dimethyl Formamide (DMF)	G	NR	E	NR
Dimethyl Sulfoxide (DMSO)	E	E	E	NR
Dicotyl Phthalate (DOP)	G	G	P	NR
Dioxane	NR	NR	NR	NR
Ethyl Acetate	F	NR	P	NR
Ethyl Alcohol	E	E	E	G
Ethylene Dichloride	NR	NR	P	NR
Ethylene Glycol	E	E	E	E
Ethyl Ether	E	E	NR	NR
Ethylene Trichloride	P	P	P	NR
Formaldehyde	E	E	E	E
Formic Acid	E	F	E	E
Freon	G	F	NR	NR
Furfural	G	NR	E	NR
Gasoline	P	E	NR	P
Gylcerine	E	E	E	E
Hexane	E	E	NR	NR
Hydraulic Fluid Petro. Based	F	E	P	G
Hydraulic Fluid Easter Based	P	P	P	P
Hydrazine 65%	E	E	G	E
Hydrochloric Acid*	G	E	E	E
Hydrofluoric Acid	G	E	E	E
Hydrogen Peroxide	E	E	E	E
Hydroquinone	G	E	E	E

Chemical	Neoprene	Nitrile	Latex	PVC
Kerosene	E	E	P	F
Lactic Acid	E	E	E	E
Lauric Acid	E	E	G	F
Linoleic Acid	E	E	P	G
Linseed Oil	E	E	P	E
Maleic Acid	E	E	P	G
Methyl Acetate	G	P	P	NR
Methyl Alcohol	E	E	E	G
Methylamine	G	E	E	E
Methyl Bromide	NR	NR	NR	NR
Methylene Chloride	NR	NR	NR	NR
Methyl Cellusolve	E	F	P	-
Methyl Ethyl Ketone (MEK)	G	NR	G	NR
Methylisobutyl Ketone	NR	P	F	NR
Methyl Methacrylate	NR	P	P	NR
Mineral Oil	E	E	P	F
Mineral Spirits	G	E	NR	F
Monoethanolamine	E	E	G	E
Morpholine	P	NR	G	NR
Muriatic Acids	E	G	G	G
Naptha V.M & P.	G	E	NR	P
Nitric Acid <30%	E	P	G	G
Nitrile Acid 70%	G	NR	F	F
Nitrile Acid Red Fuming	NR	NR	P	P
Nitrile Acid White Fuming	NR	NR	P	P
Nitrobenzene	NR	NR	P	NR
Nitromethane	E	F	G	P
Nitropropane	G	NR	E	NR
Octyl Alcohol	E	E	G	F
Oleic Acid	E	E	P	F
Paint Remover	G	G	F	P
Palmitic Acid	E	G	G	G
Pentachlorophenol	E	E	P	F
Pentane	E	E	P	NR
Perchloric Acid 60%	E	E	P	E
Potassium Hydroxide <50%*	E	G	E	E
Printing Ink	G	E	G	F
Propyl Acetate	P	F	P	NR
Propyl Alcohol	E	E	E	F
Perchloroethylene	NR	G	NR	NR
Phenol	E	NR	G	G
Phosphoric Acid*	E	E	G	G
Picric Acid	E	E	G	E
Propylene Oxide	NR	NR	P	NR
Rubber Solvent	G	E	NR	NR
Sodium Hydroxide <50%	E	G	E	G
Stoddard Solvent	E	E	P	NR
Styrene*	NR	NR	NR	NR
Sulfuric Acid 95%	F	G	NR	NR
Tannic Acid	E	E	E	E
Tetrahydrofuran (THF)	NR	NR	NR	NR
Toluene	P	G	NR	NR
Toluene Di-Isocyanate (TDI)	NR	NR	P	P
Trichlorethylene (TCE)	P	G	NR	NR
Triricrestyl Phosphate (TCP)	F	E	G	F
Triethanolamine 85% (TEA)	E	E	G	E
Tung Oil	E	E	NR	F
Turbine Oil	E	G	P	F
Turpentine	G	E	P	P
Vegetable Oil	E	E	P	F
Xylene	P	G	NR	NR

WHICH MATERIALS? (for reusable gloves)

NATURAL LATEX

Natural rubber (hevea): the most elastic of all substances known

Good resistance to wear and tearing, and to all water-soluble and diluted products. Also available in disposable gloves.

Poor resistance to oily or greasy products and to hydrocarbons. May cause allergic reactions.

NEOPRENE

Synthetic rubber based on polychloroprene

Good resistance to strong acids and bases.

Average mechanical strength Non resistant to aromatic or chlorinated solvents.

PVA

(Polyvinyl alcohol) Synthetic polymer based on vinyl alcohol

Excellent resistance to many hydrocarbons (aliphatic, aromatic and chlorinated, etc.), to esters and to ketones

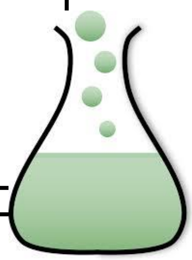
Breaks down on contact with water. Expensive

FLUORINATED MATERIALS

Synthetic materials e.g Viton® or Téflon® brands

Good resistance to many products, including benzene and PCBs

Low resistance to cuts and abrasion. Expensive



NITRILE

Synthetic rubber (acrylonitrile·butadiene·styrene copolymer or ABS)

Good mechanical strength: wide chemical resistance (oils, greases, alcohols and petroleum products, etc.). Also available in disposable gloves.

Poor resistance to ketones and halogenated products (chlorinated or fluorinated, etc.)

PVC

(Polyvinyl chloride) Synthetic polymer based on vinyl chloride, also known as "Vinyl"

Fair resistance to acids, bases and alcohols. Moderate cost. Also available in disposable gloves.

Poor resistance to ketones, aldehydes and to aromatic or halogenated hydrocarbons

BUTYL

Synthetic rubber (isobutyrene-isoprene copolymer)

Good resistance to strong acids, ketones, esters, glycol ethers, amines and aldehydes, etc.

Poor resistance to aliphatic aromatic and halogenated hydrocarbons. Expensive

MULTILAYER MATERIALS

Laminated layers of polyethylene and ethylene vinyl alcohol copolymers, brand names Barrier®, Silver Shield®

Excellent resistance to most chemicals

Reduces dexterity, poor mechanical strength

Written by G Robin
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